Name: ____________________________ Date: ________________ Period: _________

Spiral Review — Prerequisites for Lesson 3.10

1. Evaluate each inverse trigonometric expression. Give your answer in radians.

(a) $\arcsin\left(\frac{\sqrt{2}}{2}\right) =$ _______________ (b) $\arccos\left(-\frac{1}{2}\right) =$ _______________

2. Given that $\sin\theta = \frac{1}{2}$, list **all** values of θ in the interval $[0, 2\pi)$ that satisfy this equation.

$\theta =$ ______________________________

Explain briefly why there are two solutions (not just one): ______________________________

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3. Solve for $\sin x$ by isolating the trig function. **Do not** find the actual angle — just isolate $\sin x$.

$2\sin x + 1 = 0$

$\sin x =$ __________________

4. Which of the following expressions gives the value of θ in Quadrant II if $\sin\theta = \frac{3}{5}$? Circle the correct answer.

$\theta = \arcsin(\frac{3}{5})$ $\quad\theta = \pi - \arcsin(\frac{3}{5})$ $\quad\theta = \pi + \arcsin(\frac{3}{5})$